Second order homogeneous ODEs

Key Points

To solve a 2nd order differential equation of the form

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=0$$

Trial $y=e^{mx}$ for auxiliary equation

$$am^2+bm+c=0$$

- 2 real $y=Ae^{m_1x}+Be^{m_2x}$
- 1 real $y=Ae^{mx}+Bxe^{mx}$
- 2 complex $m=R\pm Ii$

$$y=e^{Rx}(\cos(Ix)+\sin(Ix))$$

Basic Skills

Q1. Solve the following quadratic equations

a) $x^2-5x+6=0$ **b)** $x^2-8x+16=0$ **c)** $x^2-8x+20=0$

Q2. Determine if the following differential equations are homogeneous and if they are constant coefficient

a) $\frac{d^2y}{dx^2}+y=0$ **b)** $\frac{d^2y}{dx^2}+y+x=0$

c) $\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=\cos x$ **d)** $x\frac{d^2y}{dx^2}+3x^2\frac{dy}{dx}+2x^3y=0$

Q3. Write the auxiliary equation for $2\frac{d^2y}{dx^2}-4\frac{dy}{dx}+y=0$

Q4. Verify that $y=3e^{2x}$ solves $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+4y=0$

Competency Skills

Q5. For the differential equation $\frac{d^2y}{dx^2}-5\frac{dy}{dx}+6y=0$

a) Write and solve the solve the auxiliary equation

b) Hence find a general solution to the differential equation

c) Given $y(0)=2$ and $\frac{dy}{dx}(0)=5$ find a particular solution

Q6. Find the general solution to the differential equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$

Q7. Find the particular solution to the differential equation $9\frac{d^2y}{dx^2}-12\frac{dy}{dx}+4y=0$ with $y(0)=0, y(1)=2$

Q8. Find the particular solution to the differential equation $\frac{2}{x}\frac{d^2y}{dx^2}+\frac{4}{x}\frac{dy}{dx}+\frac{3y}{x}=0$ given the initial conditions $y(0)=0$ and $\frac{dy}{dx}(0)=1$

Advanced Skills

Q9. A scientist has proposed a model to model the exothermic energy produced by a reaction. They say "The rate at which exothermic energy T is three times of that of the rate at which the energy is decreasing"

a) Explain why the differential equation can be written as $3\frac{d^2T}{dt}+\frac{dT}{dt}=0$

b) Given the initial energy of the system is $T=30°\mathrm{C}$ and after 2 minutes the $T=20°\mathrm{C}$ find how much exothermic energy there is after 5 minutes

c) Deduce if the system has a steady state and if so what it is

Q10. Sketch the general solution to $5\frac{d^2y}{dx^2}-2\frac{dy}{dx}+5y=0$

Extension

Q11. Write and solve a differential equation for a scenario where a car suddenly brakes and undergoes retardation by resistive forces

Hint: Consider the relationship of displacement, velocity and acceleration